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From Cherry to Cup: The Science of Coffee

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Abstract. Coffee is the most chemically studied beverage after water, yet the numbers that matter in the cup are scattered across food chemistry, agronomy, and trade literature. This paper assembles them into one quantitative account from species to serving. Green-bean chemistry differs first by species: canephora carries roughly twice the caffeine of arabica (15 to 25 versus 9 to 13 mg/g dry weight), more chlorogenic acids, and far less sucrose and oil, which explains the sensory gap between the species better than any single compound. Roasting conserves caffeine per bean while destroying most chlorogenic acids and nearly all sucrose; the "strength" of dark roast is manufactured bitterness (phenylindanes, diketopiperazines), not caffeine. Brewing parameters dominate caffeine delivery: a Vietnamese phin serving carries roughly 300 mg while a finjan of Arabic qahwa carries roughly 30 mg, a tenfold range set by dose, ratio, and bean species rather than by brew method mystique. We correct four persistent misconceptions with measurements, document 26 verified obscure, regional, and historical preparations with their extraction mechanisms, and close with an evidence-based exploration guide. All analysis was produced by Kilambi Research's quantitative research pipeline with AI assistance and verified against source data.

Keywords: coffee, caffeine, extraction yield, roasting chemistry, brewing methods, sensory science

Contents

1 Introduction **4**

2 Species, genetics, and green-bean chemistry **4**

2.1 The cultivated species 4

2.2 Varieties that matter 5

2.3 Green-bean chemistry by species 5

2.4 Growing conditions and yields 6

3 Terroir, harvest, and post-harvest processing **7**

3.1 Terroir 7

3.2 Harvesting 8

3.3 Processing methods 8

4 The chemistry of roasting **10**

4.1 Reaction networks 10

4.2 First crack, second crack, and roast development 11

4.3 Why dark roast tastes stronger without more caffeine 11

5 Brewing: mainstream methods and extraction science **13**

5.1 Two numbers that govern the cup 13

5.2 How the parameters move extraction 13

5.3 Method profiles 15

6 Obscure, regional, and historical preparations **16**

6.1 Clarified coffees: egg and shell 16

6.2 Fat, dairy, and egg enrichment 19

6.3 Salt and mineral additions 19

6.4 Spice-roasted and spiced coffees 19

6.5 Grain, root, and chicory extenders 20

6.6 Leaf and cherry teas 20

6.7 Concentrates, layers, and foam architecture 20

6.8 Fire, embers, and historical boiling 21

6.9 Animal-processed coffee 21

7 Water, grind, freshness, and sensory evaluation **22**

7.1 Water chemistry 22

7.2 Grind consistency 22

7.3 Freshness, degassing, and storage 22

7.4 Sensory science and grading 23

8 Quantitative synthesis and misconception checks **23**

8.1 Caffeine accounting 23

8.2 Four misconceptions, corrected 23

9 Discussion: falsifiability and limitations **24**

10 Conclusion: an evidence-based exploration guide **25**

A Data and methods notes **25**

B Robustness checks **25**

C Unverified leads **26**

1 Introduction

Coffee sits at an unusual intersection: a globally traded commodity of roughly 180 million bags a year, a subject of serious food chemistry, and a daily ritual shaped by vessels, habits, and local taste (International Coffee Organization, 2025). The literature on each stage of its life is deep but fragmented. Agronomists study the plant, food chemists study the roast, physicists model espresso extraction, and ethnographers record the ceremonies, and the quantitative results rarely meet in one place. The consequence is a market full of confident claims that measurements do not support: that dark roast has more caffeine, that espresso is the strongest coffee, that robusta is simply defective arabica, that cold brew is low in caffeine.

This paper builds the bean-to-cup account in numbers. Section 2 covers the biology: species, the genetics that structure them, the varieties that matter in the cup, and green-bean chemistry by species. Section 3 covers terroir, harvesting, and post-harvest processing, with what controlled studies actually show about how processing changes the beverage. Section 4 explains roasting as chemistry: Maillard reactions, caramelization, Strecker degradation, the physics of first and second crack, and the measured changes across roast levels. Section 5 compares fourteen brewing methods on common parameters and introduces the extraction science that governs them. Section 6 goes beyond the familiar methods to 26 verified obscure, regional, and historical preparations, each with its mechanism. Section 7 covers the supporting sciences: water chemistry, grind physics, freshness and storage, and sensory grading. Section 8 synthesizes the quantitative layer: caffeine accounting across species, roast, and method, and corrections to the major misconceptions. Section 9 states the paper's falsifiability conditions and limitations. Section 10 concludes with a practical exploration guide grounded in the findings.

The contribution is synthesis with teeth: every headline number carries a source and an evidence grade, disagreements between studies are reported rather than averaged away, and thin-data areas are labeled as such. Where the literature is silent, particularly on traditional brewing methods that have never seen a refractometer, we say so and estimate transparently.

2 Species, genetics, and green-bean chemistry

2.1 The cultivated species

Three species account for essentially all commercial coffee. *Coffea arabica* is the dominant species at roughly 60 to 65 percent of world production (Observed, B). *Coffea canephora*, sold as robusta, accounts for roughly 35 to 40 percent and dominates instant and soluble coffee (Observed, B). *Coffea liberica*, including the trade type Excelsa, accounts for under 1 to 2 percent of world production but dominates Malaysian plantings and is gaining attention for hot, wet lowlands where the other species struggle (Observed, A/B).

Arabica is the only allotetraploid in the genus, with 2n = 4x = 44 chromosomes, the product of a single natural hybridization between diploid *C. canephora* and *C. eugenioides* followed by genome duplication (Salojärvi et al., 2024) (Observed, A). Every other *Coffea* species is diploid at 2n = 2x = 22. The mating systems follow from the genetics: arabica is self-compatible and predominantly self-pollinating, while canephora and liberica are self-incompatible obligate outcrossers (Aerts et al., 2013) (Observed, A).

That single hybridization plus severe domestication bottlenecks left cultivated arabica with exceptionally low genetic diversity: nearly all modern cultivars descend from the Typica or Bourbon lineages founded by a handful of plants moved out of Yemen in the 17th and 18th centuries (Salojärvi et al., 2024) (Observed, A). The wild Ethiopian gene pool holds the species' remaining diversity and is the conservation priority. Canephora, as a diploid outcrosser with a large fragmented native range, retains far more diversity.

Taxonomy is currently in motion for liberica: a 2025 genomic study split the complex into three species, *C. liberica*, *C. dewevrei* (Excelsa), and *C. klainei*, with differing drought and rainfall tolerances (Nature Plants genomic study, 2025) (Observed, A). Trade usage has not caught up; "Excelsa" remains the commercial name.

2.2 Varieties that matter

The Bourbon-Typica foundation underlies nearly all traditional arabica. Typica descends from the Dutch transfer of Yemeni material to Java around 1700; Bourbon from the French planting on Ile Bourbon (Reunion) around 1720 (Salojärvi et al., 2024). Notable pure-arabica varieties include Gesha (an Ethiopian landrace whose 2004 Best of Panama win launched the modern specialty auction market, with top lots later exceeding $10,000 per kilogram), SL28 and SL34 (Kenya's blackcurrant and wine-like cup), Caturra (a 1937 Bourbon dwarf mutation), Catuai, Mundo Novo, Maragogipe, Pacamara (pure arabica despite its vigor, since both parents are arabica), Kent, and Mocha (Observed, B).

The rust-resistant families all descend from the Timor Hybrid, a spontaneous arabica-by-canephora cross found on Timor in the 1920s to 1940s: Catimor, Sarchimor, Colombia, Castillo, Ruiru 11, and Batian (Observed, B). The breeding literature records the central tradeoff plainly: the canephora-derived traits conferring pest resistance are partly the same traits that depress cup quality, so introgressed cultivars sit on a resistance-quality frontier rather than escaping it (Farah, 2012) (Observed, A). A controlled sensory trial found no significant quality difference between Castillo and Caturra grown and processed alike, confirming that the resistance need not cost the cup when the rest is held constant (Observed, A/B).

2.3 Green-bean chemistry by species

Table 1 gives the reference composition. The pattern is stark and explains most of the sensory gap between the species. Canephora green beans contain approximately twice the caffeine of arabica: 15 to 25 mg/g versus 9 to 13 mg/g dry weight (Farah, 2012) (Observed, A), a ratio corroborated by independent studies at roughly 1.8 to 2.2 times. Caffeine alone cannot reliably discriminate species in all samples, with overlapping ranges in about 4 percent of samples in one study, but the bands are distinct (Observed, A).

Arabica's advantages as a beverage raw material are chemical: roughly double the sucrose (6.0 to 9.0 percent versus 0.9 to 4.0 percent, subject to the dispute noted), roughly double the lipids (15 to 17 percent versus 7 to 10 percent), and lower chlorogenic acids (4.1 to 7.9 percent versus 6.1 to 11.3 percent) (Farah, 2012) (Observed, A). Sucrose is the principal flavor precursor for caramelization and Maillard reactions; lipids carry aroma and build mouthfeel. Canephora's higher chlorogenic acids protect the plant but oxidize into cup defects, which is the chemical face of the resistance-quality tradeoff. Liberica and Excelsa sit intermediate to arabica-like in caffeine, but the

Table 1. Green-bean chemistry by species, percent of dry weight

Compound	*C. arabica*	*C. canephora*
Caffeine	0.9–1.3	1.5–2.5
Chlorogenic acids	4.1–7.9	6.1–11.3
Lipids (coffee oil)	15–17	7.0–10.0
Sucrose	6.0–9.0	0.9–4.0[a]
Trigonelline	0.6–2.0	0.6–0.7
Protein/peptides	10.0–11.0	11.0–15.0

Notes: Primary source is Farah (2012), Table 2.1. Figure 2 plots midpoints. [a]The canephora sucrose band is disputed: Brunning (2023) gives 3.0–7.0 percent from the same literature base; treat robusta sucrose as approximately 1–7 percent pending primary confirmation.
Source: Farah (2012), with corroboration noted in text.

literature disagrees on the band (1.2 to 3.8 g/100 g in a 2026 review versus 0.89 percent for excelsa in a Comorian metabolomics study); the honest statement is overlap with arabica and generally below robusta (RECONSTRUCTED, B).

Figure 1 shows the caffeine bands. The "roughly double" rule survives across sources, and it is the single most load-bearing number in the paper: nearly every caffeine comparison downstream starts here.

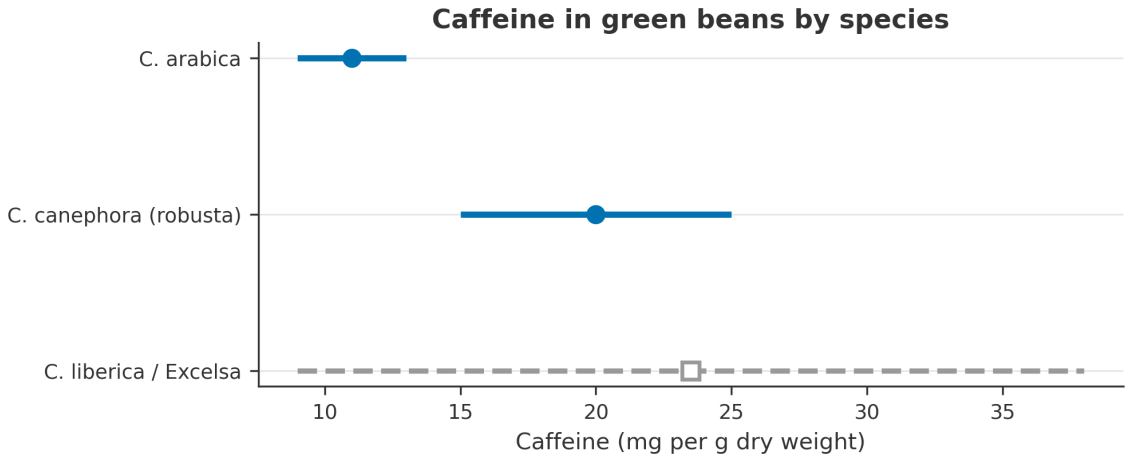


Figure 1. Caffeine in green beans by species. Bars span reported ranges, markers show midpoints. Liberica/Excelsa is dashed because the literature disagrees on its band.
Notes: Arabica and canephora bands from Farah (2012), Table 2.1, corroborated by independent studies. Liberica/Excelsa band is thin-data: a 2026 review reports 1.2–3.8 g/100 g while a Comorian metabolomics study measured 0.89 percent for excelsa.
Source: Research notes, Section 2, data through October 2026.

2.4 Growing conditions and yields

Arabica wants cool highlands: optimal mean temperatures of roughly 18 to 21 C, commercial altitudes of 600 to 2,200 m, rainfall 1,200 to 2,200 mm per year (OBSERVED, A/B). Canephora

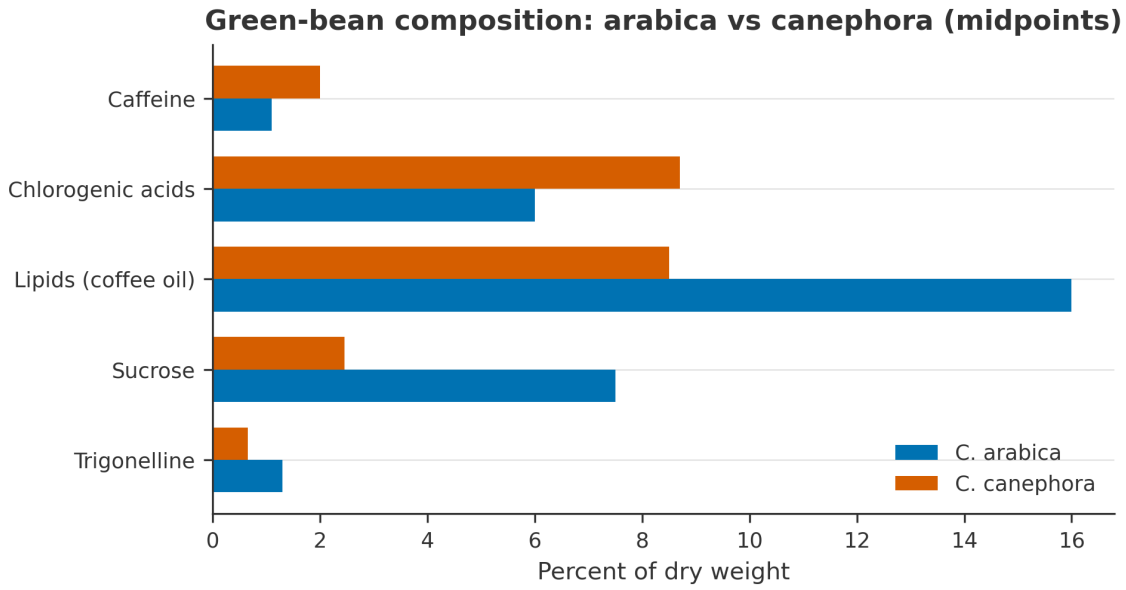


Figure 2. Green-bean composition: arabica versus canephora. Bars show midpoints of the bands in Table 1.
Notes: Midpoints of Farah (2012) bands. The canephora sucrose bar uses Farah's band; Brunning (2023) reports 3.0–7.0 percent. See Table 1 note.
Source: Farah (2012), via research notes, data through October 2026.

wants hot lowlands: 0 to 800 m, warmer and wetter. Liberica and Excelsa tolerate hot, wet lowlands to about 1,200 m, poor soils, and seasonal drought (particularly dewevrei), which is their agronomic asset under climate change (Nature Plants genomic study, 2025) (Observed, A/B).

Yields vary enormously with inputs and biennial bearing, so only contextual ranges are meaningful. Brazil, the most input-intensive large producer, averaged a record 34.6 bags per hectare (about 2,076 kg of green coffee per hectare) in 2026 (Conab, 2026) (Observed, A). Indonesian robusta smallholders average 600 to 800 kg/ha against 1,500 to 2,000 kg/ha achievable with improved practice (USDA Foreign Agricultural Service, 2025) (Observed, A). Typical African and Asian smallholder arabica runs 400 to 800 kg/ha (Observed, B).

3 Terroir, harvest, and post-harvest processing

3.1 Terroir

Altitude is the most studied terroir variable because it acts through a clean mechanism: cooler temperatures at elevation slow cherry maturation, extending the bean-filling period. An Ethiopian study across 20 zones found sucrose, fat, trigonelline, and bean weight rising with elevation while chlorogenic acids fell, with rainfall not significant (Urugo et al., 2024) (Observed, A). A factorial trial in Mozambique measured roughly four extra weeks of maturation across a 650 to 935 m gradient (Observed, B). Specialty quality concentrates above 1,000 m for arabica, with exceptional lots above 1,700 m; the Gesha variety expresses its floral profile most fully above 1,700 m in Panama (Observed, B).

Shade versus sun is secondary to altitude in the evidence: shade delays ripening similarly but carries

roughly a 20 percent yield penalty, and its cup effects are inconsistent across studies (Observed, B). Soil pH targets run 4.5 to 6; mineral effects on the cup are real but small relative to altitude and processing (Observed, B).

3.2 Harvesting

Selective hand-picking of ripe cherry is the quality standard because cherry composition changes during maturation: sugars in the coat and juice roughly double while endosperm sucrose stays flat, so underripe and overripe cherry dilute the lot (Goncalves et al., 2024) (Observed, A). Strip picking and mechanical harvesting trade labor cost for a wider ripeness spread and up to roughly 20 percent defect mass before sorting (Observed, B). A Colombian maturity-composition cup trial linked maturity indices to cup score with R-squared of 0.83 to 0.85 (Observed, B).

3.3 Processing methods

Processing is controlled fermentation and drying of the cherry around the seed, and it is the largest post-harvest lever on flavor. Figure 3 summarizes the directional effects; the scores are reconstructed from qualitative trade consensus and controlled trials (graded C, see notes).

Washed (wet) process. Cherry is pulped, the mucilage fermented off (12 to 36 hours) or removed mechanically, then the parchment dried. The cup is clean, bright, high-acid, light to medium in body, with floral and citrus aromatics. A controlled metabolomics trial scored washed at 79.50 against natural 77.75 and honey 77.25 on the same lot, confirming the direction if not the universality of the ranking (Observed, B). Water use is the cost: 0 to 100 cubic meters per tonne of green coffee, mean 22.6 in a 2024 life-cycle assessment, against near zero for naturals (Observed, B).

Natural (dry) process. Whole cherry dries in the fruit, typically 2 to 4 weeks on raised beds or patios. Fruit sugars and fermentation metabolites migrate into the seed. The cup is sweet, heavy-bodied, lower in perceived acidity, with berry, tropical fruit, and winey notes and high fermentation character. Risks are mold and phenolic taint if drying stalls.

Honey process. Pulped cherry dried with part or all of the mucilage intact; yellow, red, and black denote decreasing mucilage removal and increasing drying time. The cup runs between washed and natural: medium-high acidity and sweetness, medium to full body, honeyed and jammy notes. Black honeys approach naturals in intensity and risk.

Wet-hulled (giling basah). Sumatran practice: parchment is hulled at roughly 30 to 50 percent moisture, far wetter than the usual 10 to 12 percent, then dried as green bean. The cup is very low in acidity, very heavy in body, earthy, herbal, tobacco and spice notes. The mechanism is trade consensus plus one storage trial rather than controlled chemistry (Reconstructed, C); mold and ochratoxin risks are the documented costs.

Anaerobic fermentation. Cherry or pulped coffee ferments in sealed tanks, sometimes with CO2 flushing, for 24 to 96 hours or more. The cup shows intense fruit, wine-like and floral aromatics, rounded acidity, high sweetness. The main failure mode is acetic, vinegar character when temperature runs hot or oxygen leaks in (Observed, B).

Carbonic maceration. Borrowed from winemaking: whole cherry ferments under CO2. One study moved cup scores from 80.90 to 85.15 with bright fruit, spice, and complexity, but defects rise after about 10 days and temperature control is mandatory (Observed, B).

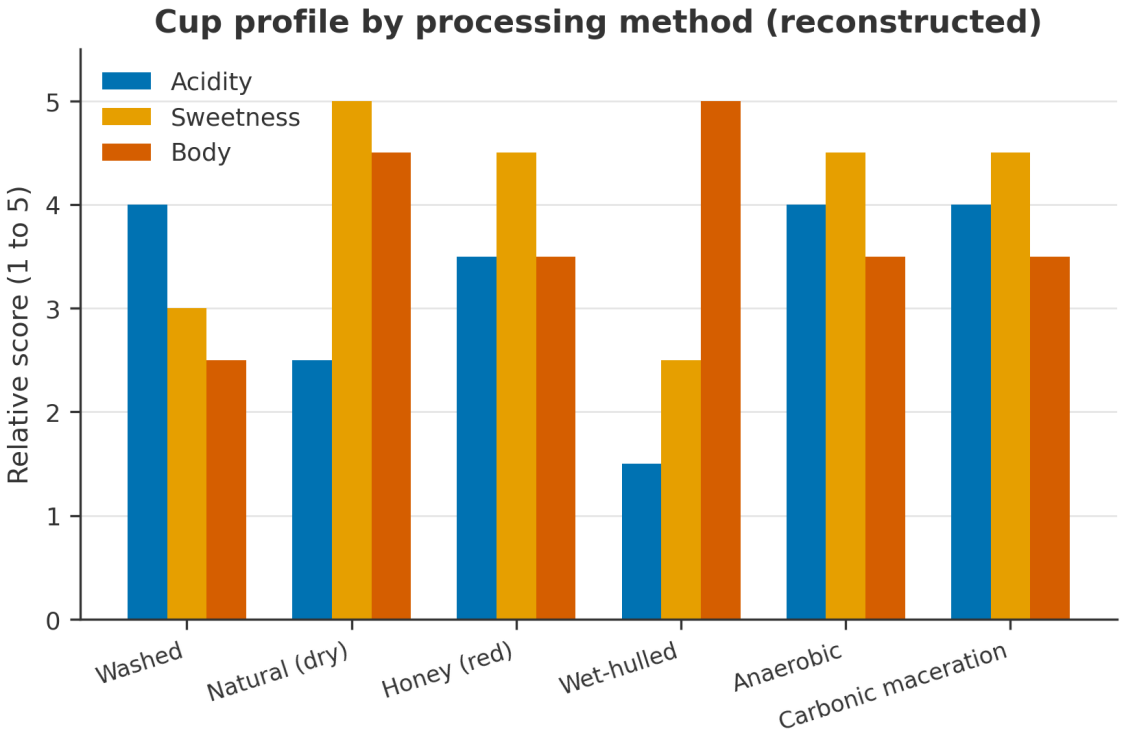


Figure 3. Cup profile by processing method. Scores are reconstructed on a 1 to 5 scale from qualitative consensus and controlled trials.

Notes: Reconstructed (Reconstructed, C) from the comparison table in research notes, Section 3. Washed/natural/honey directionality is (Observed, B) (multi-study agreement plus one controlled metabolomics trial); anaerobic and carbonic maceration rows rest on fewer, smaller studies; the wet-hulled row is trade consensus plus one storage trial.

Source: Research notes, Section 3, data through October 2026.

Yeast-inoculated and experimental fermentations. Inoculation with *Saccharomyces cerevisiae* or *Torulaspora delbrueckii* gives controlled, strain-dependent ester profiles and aromatic lift; wrong strain or dose gives off-flavors (Observed, B).

Drying discipline underlies all of it: thin layers, frequent turning, raised beds for airflow. The safe target is below 12.5 percent moisture (water activity 0.67 to 0.70) per FAO/WHO Codex CXC 69-2009 (Observed, A); commercial lots in a Brazilian multi-method study ran 7.7 to 10.4 percent with water activity below 0.6 (Observed, A). Whole-cherry drying takes roughly 12 days through three microbial phases versus about 7 days after wet processing, which is why wet-processed coffee carries lower ochratoxin risk (Observed, B).

One honest gap: controlled beverage-pH-by-process data are sparse in the literature. Acidity differences between processes are well supported directionally by cupping and metabolomics, but the paper does not report a pH table the evidence cannot fill.

4 The chemistry of roasting

4.1 Reaction networks

Green coffee is a dense seed at 8 to 12 percent moisture with almost none of the beverage's flavor. Roasting applies heat until internal bean temperatures reach roughly 190 to 240 C, driving off 12 to 20 percent of mass while the bean expands substantially in volume (50 to 80 percent in trade measurements; the direction is confirmed in peer-reviewed work but the band is trade literature) (Illy and Viani, 1995) (Assumed, C). Three reaction families do most of the work.

Maillard reactions. Nonenzymatic browning between reducing sugars and amino acids, significant from about 140 to 150 C bean temperature. The cascade runs through Amadori rearrangement to reductones, furans, and finally melanoidins, the brown nitrogenous polymers that make up roughly 25 to 29 percent of roasted dry matter (Hofmann, 1998) (Observed, A). Maillard chemistry is the primary generator of roast aroma: pyrazines (nutty, roasty), furans and furanones (caramel, sweet), pyrroles, and thiols (Flament, 2002) (Observed, A).

Strecker degradation. Amino acids react with Maillard dicarbonyls to give Strecker aldehydes (leucine to 3-methylbutanal, malty; phenylalanine to phenylacetaldehyde, floral-honey) plus CO2 and aminoketones that feed pyrazine formation (Poisson et al., 2020) (Observed, A). Each turnover releases one CO2 molecule, and Strecker degradation accounts for 80 percent or more of the CO2 formed inside the bean during roasting (Reconstructed, B); that CO2 is what pressurizes the bean and ultimately drives second crack.

Caramelization. Thermal degradation of sugars without amino acids, mostly the story of sucrose. Sucrose falls 47 percent by mid-roast at medium level and 59 percent at full city, with 96 to 98 percent gone by the end of the roast (Divis et al., 2018) (Observed, A). The caramel note of a medium roast comes from furanones and maltol formed here, not from residual sugar, because essentially no sucrose survives a developed roast.

Chlorogenic acids degrade in parallel: about 60 percent lost under mild roasting, approaching 100 percent under severe roasting, with isomer-specific rates (Trugo and Macrae, 1984) (Observed, A). Early in the roast, CGA lactones form and give the mild pleasant bitterness of light to medium cups; continued roasting converts them to phenylindanes, the harsh lingering bitterness of dark roasts (Frank et al., 2006) (Observed, A). Caffeine contributes only about 15 percent of perceived

bitterness (Observed, A). Trigonelline degrades 50 to 90 percent to nicotinic acid and pyridines (Observed, B).

Total volatile concentration does not rise monotonically: light roasts preserve fruity esters, medium roasts maximize total volatile abundance, and dark roasts trade complexity for pyridines, pyrazines, and phenols (Observed, A/B).

4.2 First crack, second crack, and roast development

First crack at 196 to 199 C bean temperature is steam-pressure rupture of cells: water vaporizes inside a still-strong cell wall matrix until it bursts, opening the pore structure (Yergenson and Aston, 2019) (Observed, A). Second crack at roughly 224 to 230 C is CO2-pressure fracture of the heat-weakened cellulose matrix; oil migrates through the fractured walls to the surface, which is why beans turn visibly oily after second crack (Observed, A/B).

Roasters divide the curve into drying, Maillard (browning), and development phases, the last measured from first crack to drop. Roast degree is standardized by color: the Agtron/SCA system runs from tile 95 (very light) to 25 (very dark), with the SCAA cupping protocol targeting about 58 whole bean (Observed, B). We use light, medium, medium-dark, and dark as operational bands anchored to mass loss and crack events (Table 2).

Table 2. Roast level and measured changes (arabica drum roasts, typical ranges)

Attribute	Light	Medium	Med-dark	Dark
Drop point	after 1st crack	between cracks	into 2nd crack	past 2nd crack
Mass loss (% of green)	12–14	14–17	17–19	18–22
CGA retained (% of green)	40–55	30–55	5–20	0–5
Sucrose retained (% of green)	20–40	5–20	1–5	0–2
Caffeine per bean vs green	95–100	93–100	90–98	90–95
Caffeine per gram vs green	100–105	103–110	108–115	110–118
Acidity	high, bright	high, balanced	moderate	low
Bitterness	low-mild	moderate	high	very high
Body	light	medium	med-full	full
Surface oil	none	none	spotting	oily

Notes: CGA bands span Trugo and Macrae (1984), Divis et al. (2018), and a multi-origin CQA study whose roast definitions differ; treat as indicative. Caffeine-per-gram rises are arithmetic consequences of mass loss. Figure 4 plots mass loss, CGA retention, and the caffeine index.
Source: Research notes, Section 4; Illy and Viani (1995), Trugo and Macrae (1984), Divis et al. (2018).

4.3 Why dark roast tastes stronger without more caffeine

Three facts resolve the confusion. Per bean, caffeine is nearly constant across roast levels (losses of roughly 5 to 10 percent, physical not chemical) (Rao et al., 2024) (Observed, A). Per gram, caffeine concentration rises slightly with roast purely as a mass-loss artifact (about 16 percent from green to French roast in one multi-origin study) (Observed, C, mechanism A). Per scoop, light roast delivers slightly more because dark beans are less dense (Reconstructed, B/C). The

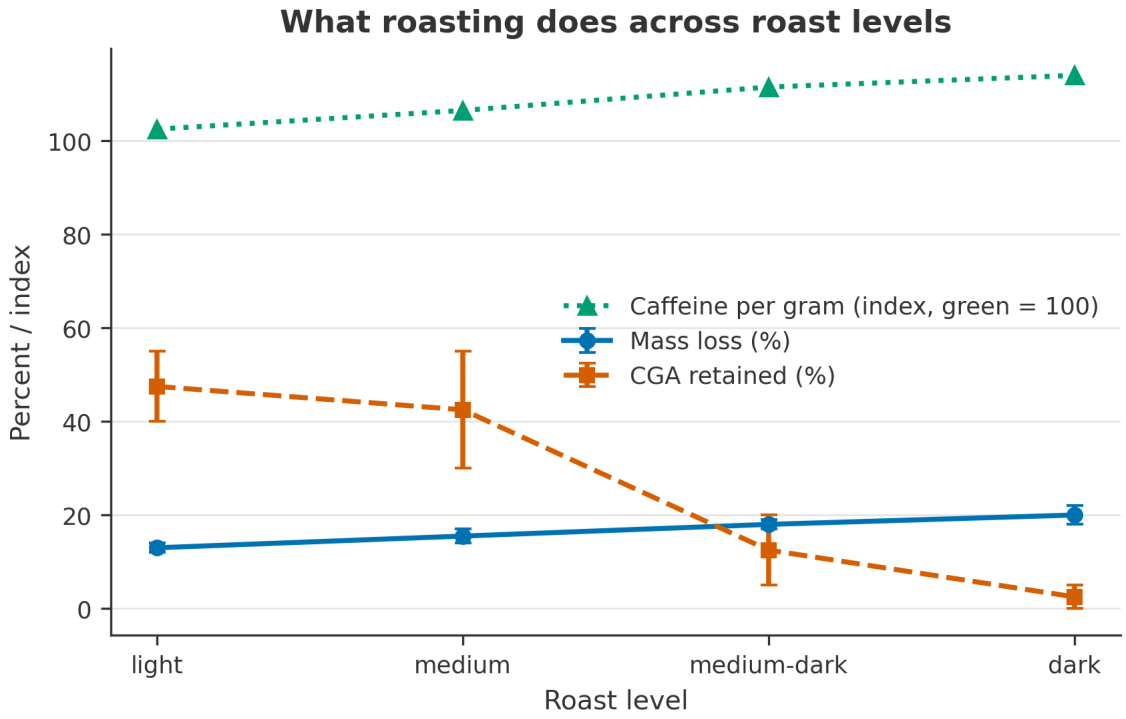


Figure 4. Roasting across roast levels: mass is lost, chlorogenic acids are destroyed, caffeine per gram rises only because the bean gets lighter. Markers show midpoints, bars the reported bands.

Notes: Mass loss and CGA retention from the studies in Table 2. Caffeine per gram (index, green = 100) is arithmetic: caffeine is conserved while 12 to 22 percent of bean mass is lost. Caffeine is thermally stable at roasting temperatures; per-bean losses are roughly 5 to 10 percent via sublimation and steam entrainment (Rao et al., 2024).

Source: Research notes, Section 4, data through October 2026.

best-controlled study, roasting one Ethiopian coffee to five degrees and brewing at fixed ratio with sieved grinds, found cup caffeine peaking at light-to-medium roast (14 to 16 percent mass loss) and falling at both extremes, with extraction yield peaking at 12 to 14 percent mass loss (Rao et al., 2024) (Observed, A).

Dark roast tastes stronger for three roast-driven reasons: bitterness rises monotonically as lactones convert to phenylindanes and diketopiperazines accumulate; body rises as melanoidins accumulate and surface lipids migrate outward; and roast flavors (guaiacol, pyridines, carbonized notes) mask origin character and read as intensity (Observed, A/B). None of these is caffeine.

5 Brewing: mainstream methods and extraction science

5.1 Two numbers that govern the cup

Two measurements describe any brew. Total dissolved solids (TDS) is the mass percent of dissolved coffee in the beverage: strength. Extraction yield (EY) is the mass of dissolved solids divided by the dry coffee dose: how much of the bean made it into the cup. The industry's balanced window is roughly 18 to 22 percent EY; below 16 percent tastes sour and thin, above 24 percent bitter and harsh, though the window moves with coffee and method (Observed, B).

Figure 5 plots the methods that have published refractometer data on these axes. Espresso lives far up the strength axis (8 to 12 percent TDS) at normal extraction; moka pot reaches the highest extraction yields measured (27.6 to 31.9 percent) because near-boiling water contacts the bed for minutes (Navarini et al., 2009) (Observed, A); the filter methods cluster in the 1.2 to 1.6 percent TDS band. The traditional methods in Tables 3 and 4 marked n.d. have never been instrumented in the published literature, which is itself a finding: the methods with the longest histories are the least measured.

5.2 How the parameters move extraction

Grind size. Finer grinding increases surface area and shortens diffusion paths, accelerating extraction of all compounds. Particles below about 100 microns ("fines") extract disproportionately and clog flow paths; grinding finer is the largest single lever on speed, until it backfires through channeling (Observed, B).

Temperature. Higher temperature accelerates diffusion and dissolves the less-soluble bitter compounds (melanoidins, chlorogenic acid derivatives). The standard 90 to 96 C window is a compromise: hot enough for sugars and acids, cool enough to limit harshness. Cold brewing inverts the trade: 12 to 24 hours at 4 to 22 C reaches comparable caffeine with lower acid and bitter extraction (Wang and Lim, 2023) (Observed, A).

Contact time. Extraction follows a saturating curve: more than 70 percent of espresso antioxidants wash out in the first 8 seconds (Ludwig et al., 2014) (Observed, A). Extending a shot past about 30 seconds adds mostly bitter compounds, not caffeine. This is why "pulling longer adds proportional caffeine" is false.

Pressure. In espresso, about 9 bar permeates water through a tamped fine bed and emulsifies oils and CO2 into crema; pressure does not much change solubility, its role is mechanical (Observed,

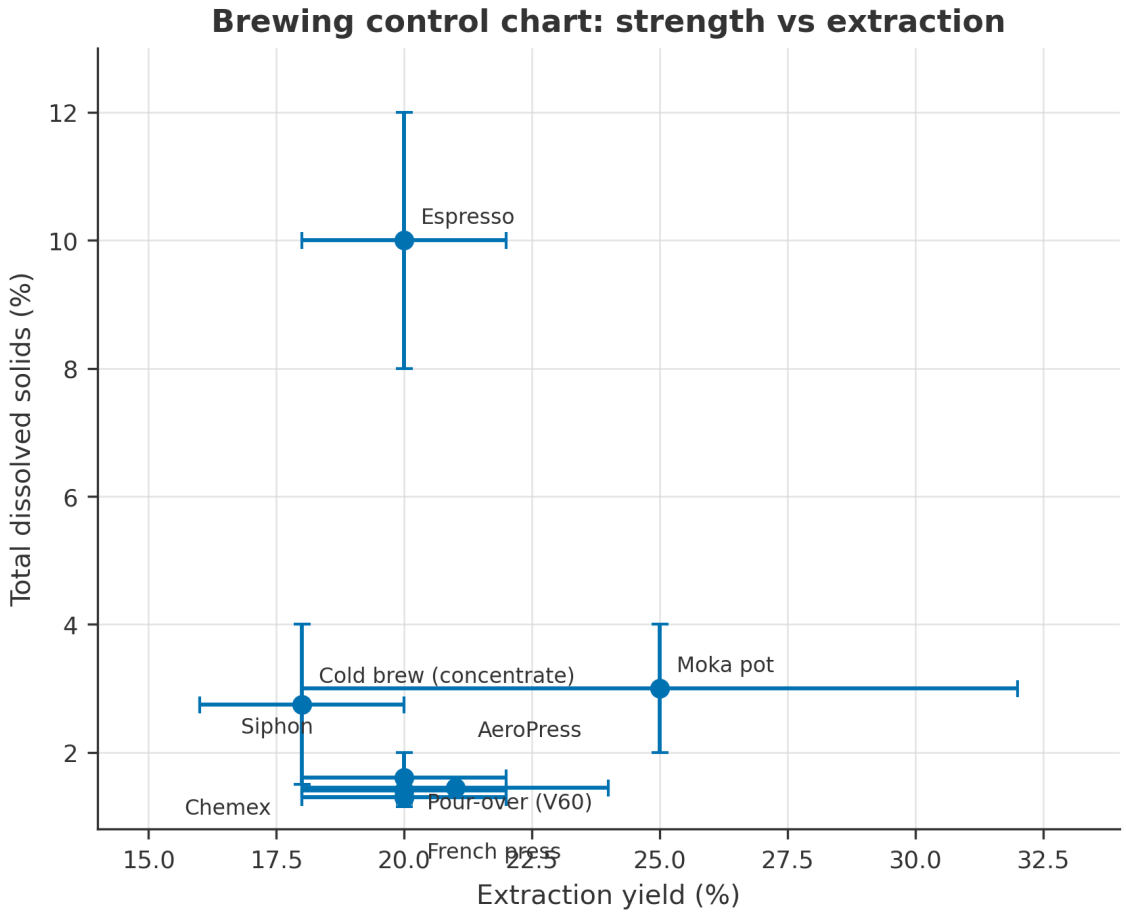


Figure 5. Brewing control chart. Markers show midpoints, bars the reported ranges. Strength (TDS) and extraction (EY) are independent: espresso is strong at normal extraction, moka extracts the most, cold brew concentrate is strong at modest extraction.

Notes: Ranges from SCA gold-cup windows, Navarini et al. (2009), Gloess et al. (2013), and Wang and Lim (2023). Traditional methods (Turkish, qahwa, jebena, phin, South Indian filter, Kyoto) lack published TDS/EY data and are excluded.

Source: Research notes, Section 5, data through October 2026.

B). Crema acts as an aroma-sealing lid, retaining low-volatility compounds while intensifying high-volatility ones in the headspace (Gloess et al., 2013) (OBSERVED, A).

Brew ratio. Less water per gram raises TDS roughly in inverse proportion in immersion brewing. Ratio is the dominant control on both strength and total caffeine per serving: a 1:8 cold brew concentrate is about twice the TDS of a 1:16 pour-over at similar extraction (OBSERVED, B).

Channeling. Water follows the path of least resistance, and small pore variations have outsized effects: doubling a pore's diameter increases flow through it sixteenfold. Hendon et al. (2020) modeled espresso as two regions of differing packing density and showed a positive feedback loop in which more flow causes more extraction, lowering resistance and widening the disparity; the confirmed prediction is that grinding finer can produce weaker, less even espresso (OBSERVED, A). Uniform particle size distribution matters more than absolute fineness.

5.3 Method profiles

Tables 3 and 4 put fourteen methods on common footing. The notes below highlight what each method's physics implies.

Espresso. 18 to 20 g dose, 36 to 40 g beverage (1:2), 25 to 30 seconds at 92 to 94 C and about 9 bar; 8 to 12 percent TDS at 18 to 22 percent EY. A double carries 120 to 190 mg of caffeine; the classic Italian single (7 to 9 g) about 60 to 75 mg (OBSERVED, B). Ludwig et al. (2014) measured 174 mg in 22 ml of espresso against 33 mg in 22 ml of filter coffee from the same beans: per unit volume, espresso is the most concentrated method by far (OBSERVED, A).

Moka pot. Steam pressure of order 1 to 2 bar drives water up through the bed. Extraction begins well below 100 C, contrary to the common belief; the cycle ends in a "strombolian phase" where superheated steam transits the bed and solubilizes bitter, astringent, smoky compounds (Navarini et al., 2009) (OBSERVED, A). It is the most extraction-efficient mainstream method (Gloess et al., 2013), which is why it is also the easiest to push bitter.

French press. Full immersion, coarse grind, 4 minutes at 92 to 96 C, metal mesh filtration. No percolation gradient, so extraction is uniform across the bed; the mesh passes oils and fines, giving heavy body and low clarity. A 240 ml cup carries roughly 150 to 170 mg of caffeine (RECONSTRUCTED, B).

Pour-over (V60) and Chemex. Gravity percolation through paper. The V60's thinner filter and the Chemex's filter (roughly three times the paper mass) bracket the clarity range: the Chemex cup is exceptionally clean and tea-like because lipid-soluble aromatics are stripped. Both sit at 18 to 22 percent EY when dialed in; shallow beds at small doses extract unevenly and run bitter from excess fines (OBSERVED, B).

AeroPress. Immersion plus manual pressure (about 0.5 bar) for fast, clean separation. Competition recipes routinely use 80 to 86 C water with extended steeps, demonstrating that time substitutes for temperature (OBSERVED, B).

Turkish coffee. Powder-ground (under 200 microns, finer than espresso) boiled with water, served unfiltered so grounds settle in the cup. Extraction efficiency is high from powder fineness plus boiling, though published EY figures do not exist. A 60 ml cup carries roughly 50 to 65 mg (RECONSTRUCTED, B); foam is the quality marker.

Arabic qahwa. Lightly roasted ("blonde") arabica simmered in a dallah with cardamom (often one part to three of coffee), served unsweetened in small finjan cups with dates. Dilute ratio and small serving give roughly 20 to 35 mg per 60 ml finjan (ASSUMED, C); no published TDS or EY data exist.

Ethiopian jebena. Fresh-roasted beans pounded in a mortar, boiled in a clay jebena, poured from height; three successive brews (abol, tona, baraka) from the same grounds. Repeated boiling extracts progressively more bitter compounds while caffeine and aromatics deplete across rounds; abol carries roughly 35 to 50 mg per 60 ml cup (ASSUMED, C).

Vietnamese phin. Gravity drip through a perforated plate at 1:8 to 1:10, traditionally robusta, over sweetened condensed milk. The arithmetic is stark: 18 g at 2.2 percent caffeine is 396 mg in the dose, and roughly 280 to 330 mg reaches the 120 ml serving (RECONSTRUCTED, C), among the highest single servings in the tables, usually diluted with milk and ice.

South Indian filter coffee. Dark roast blended with 10 to 30 percent roasted chicory, slow-percolated into a concentrated decoction, mixed 1:2 to 1:3 with boiled full-fat milk and aerated between dabara and tumbler. Chicory slows percolation, adds body, and dilutes caffeine (chicory is caffeine-free); a tumbler carries roughly 50 to 70 mg (ASSUMED, C).

Siphon. Vapor pressure lifts near-boiling water into an upper chamber for immersion, then vacuum draws the brew through cloth. The sealed hot environment gives very even extraction; cloth filtration gives exceptional clarity. Measured TDS reached 2.02 percent, the highest among gravity methods tested in one comparison (OBSERVED, B).

Cold brew. Coarse coffee steeped 12 to 24 hours at 4 to 22 C; time substitutes for heat. Wang and Lim (2023) found extraction rate and yield rising with temperature and falling grind size, with the best sensory condition (1:10, 16 hours, 500 microns, 4 C) at 1.42 percent TDS and 2.31 g/L caffeine; cold brews showed higher pH and lower titratable acidity than hot French press (OBSERVED, A). "Cold brew has less caffeine" is false: a 240 ml glass diluted 1:1 from a 1:8 concentrate carries roughly 150 to 250 mg (RECONSTRUCTED, B). The ratio, not the temperature, dominates.

Kyoto drip. Ice water drips at about one drop per 1.5 to 3 seconds through medium-coarse coffee over 3 to 5 hours at roughly 1:6, producing a viscous concentrate served in 30 to 60 ml portions. No published TDS, EY, or caffeine data exist (ASSUMED, C).

6 Obscure, regional, and historical preparations

Beyond the fourteen methods above, coffee is prepared in dozens of local ways that rarely appear in the brewing literature. We verified 26 against named sources, each rated High, Medium, or Low confidence; six further leads could not be verified and are quarantined in Appendix C. For each method we report the mechanism by which it changes the cup relative to conventional brewing. Table 5 summarizes all 26.

6.1 Clarified coffees: egg and shell

Scandinavian-American egg coffee. Coarse grounds are boiled with a whole beaten egg (sometimes with the crushed shell), then cold water is added to settle everything. Egg-white albumin coagulates in the hot brew and binds fines and tannins, clarifying the cup much as egg whites clarify consomme;

Table 3. Brewing parameters by method

Method	Ratio	Grind	Temp (C)	Pressure	Contact time	Filtration
Espresso (double)	1:2	fine	92–94	9 bar	25–30 s	metal
Moka pot	∼1:10	fine-med	70–96	1–2 bar	2–4 min	metal plate
French press	1:15	coarse	92–96	none	4 min	metal mesh
Pour-over (V60)	1:15–1:17	med-fine	90–96	none	2.5–3.5 min	paper
Chemex	1:16	med-coarse	90–96	none	4–5 min	thick paper
AeroPress	1:12–1:15	fine-med	80–96	∼0.5 bar	1–2 min	paper/metal
Turkish	1:8–1:10	powder	to boil	none	3–5 min	none
Arabic qahwa	∼1:30	med-fine	simmer	none	10–15 min	settle
Ethiopian jebena	∼1:12	mortar	boil	none	∼5 min	none
Vietnamese phin	1:8–1:10	med-fine	92–96	none	4–8 min	metal plate
S. Indian filter	decoction	fine	∼95	none	10–30 min	metal
Siphon	1:15	medium	92–96	vapor/vac	1–2 min	cloth/paper
Cold brew	1:8	coarse	4–22	none	12–24 h	paper/metal
Kyoto drip	∼1:6	med-coarse	0–4	none	3–5 h	paper

Notes: Standard preparations; see text for dose and beverage weights. "n.d." in Table 4 marks cells with no published measurement.
Source: Research notes, Section 5.

Table 4. Brewing outcomes by method

Method	TDS (%)	EY (%)	Caffeine per serving (mg)	Body / clarity
Espresso (double)	8–12	18–22	120–190	full / medium
Moka pot	2–4	18–32	110–150	full / med-low
French press	1.35–1.45	18–22	∼150–170	heavy / low
Pour-over (V60)	1.15–1.45	18–22	∼180–210	light-med / high
Chemex	1.2–1.4	18–22	∼140–170	light / very high
AeroPress	1.3–1.6	18–24	∼150–190	medium / high
Turkish	n.d. (high)	n.d.	∼50–65	very heavy / low
Arabic qahwa	n.d.	n.d.	∼20–35[a]	light-med / medium
Ethiopian jebena	n.d.	n.d.	∼35–50[a]	med-heavy / low
Vietnamese phin	∼3–5[a]	n.d.	∼280–330[a]	heavy / medium
S. Indian filter	n.d.	n.d.	∼50–70[a]	heavy / med-low
Siphon	1.2–2.0	18–22	∼180–220	medium / very high
Cold brew	1.5–4[b]	16–20	∼150–250	med-full / medium
Kyoto drip	n.d.	n.d.	n.d.	heavy / high

Notes: Servings are standard preparations (see text). [a]Estimated (ASSUMED, C): no published refractometer data exist for these traditional methods. [b]Concentrate before dilution. Figure 6 plots the caffeine column.
Source: Research notes, Section 5; Depaula and Farah (2019), Ludwig et al. (2014), Mystkowska et al. (2024), Wang and Lim (2023), Navarini et al. (2009).

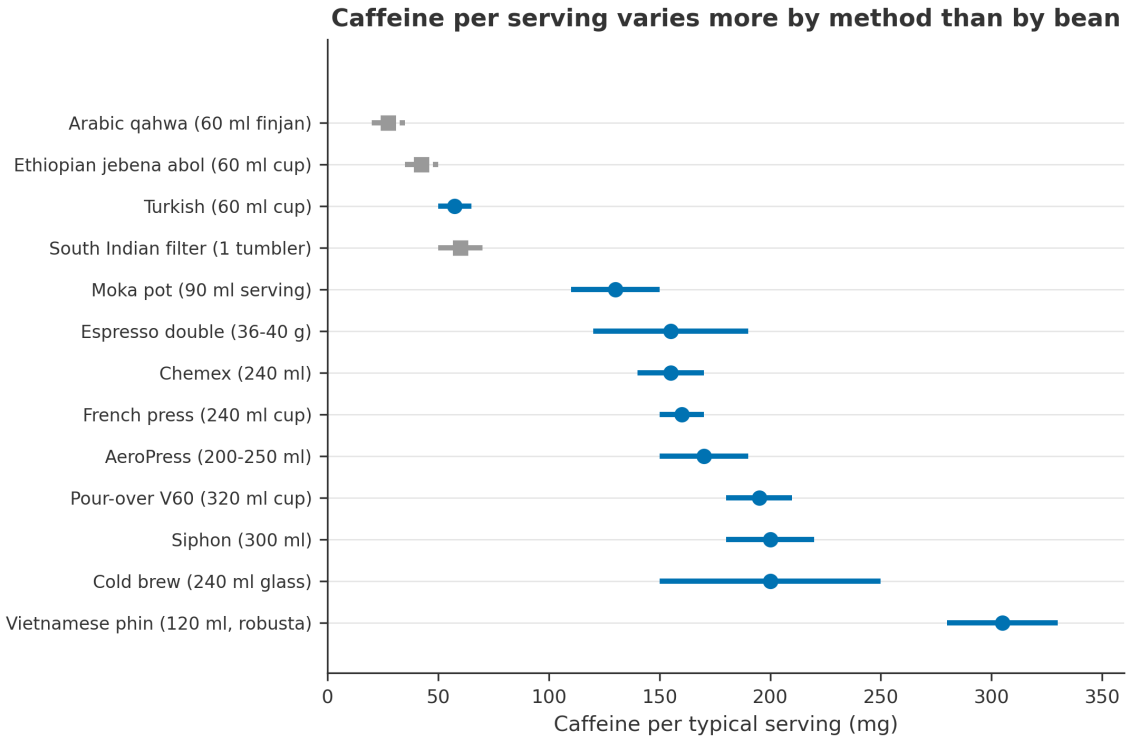


Figure 6. Caffeine per typical serving by brewing method. Markers show midpoints, bars the reported or estimated ranges. Gray dashed bars are estimated where no published measurement exists.

Notes: Values from Table 4. The tenfold range from qahwa (∼30 mg) to phin (∼300 mg) is set by dose, brew ratio, and species, not by method sophistication.

Source: Research notes, Section 5, data through October 2026.

shell calcium carbonate buffers acidity (Christensen, n.d.) (Observed, B). The result is a clear, low-bitterness cup from a boil that would otherwise be harsh. The method is High confidence as a technique, though the name misleads: it is virtually unknown in Scandinavia today and appears to be a 19th-century Scandinavian-American immigrant invention, not a Scandinavian tradition (Observed, B).

6.2 Fat, dairy, and egg enrichment

Fat changes coffee in two ways: lipids carry fat-soluble aroma compounds and round mouthfeel, and dairy proteins bind bitter polyphenols.

Ca phe trung (Vietnam, Hanoi). Whipped egg yolk with sweetened condensed milk is frothed over strong robusta, served hot or iced. The yolk-milk foam is custard-like, tiramisu in a cup, over a bitter robusta base; the fat and sugar mask robusta's harshness while the coffee supplies the bitter counterpoint (Observed, B).

Kaffeost (Lapland/Finland). Cubes of leipajuusto ("bread cheese") sit under hot coffee. A two-phase cup: the cheese softens, dairy fat and protein round bitterness, and the drinker eats the coffee-soaked cheese (Observed, B/C).

Hainanese kopi (Malaysia/Singapore). Beans are wok-roasted with margarine and sugar, then brewed through a muslin "sock" with a high pour. The fat-roast caramelizes sugars onto the bean and adds buttery, caramelized notes; the sock passes oils for a full-bodied cup (Observed, B). This is roasting *with* fat, distinct from adding it later.

Ethiopian ceremony variants. Rural practice adds salt or niter kibbeh (spiced clarified butter) to jebena coffee; no source describes a standalone butter-coffee preparation, so this remains a variant note (Observed, C).

6.3 Salt and mineral additions

Ca phe muoi (Vietnam, Hue). Salted whipped cream over iced robusta. Salt suppresses bitterness perception at the receptor level, producing a salted-caramel rounding of an otherwise aggressive cup (Observed, B). The single-cafe origin story (a Hue cafe in 2010) rests on one travel guide's reporting (Observed, C).

Kopi joss (Yogyakarta, Indonesia). A piece of red-hot charcoal is plunged into sweetened black coffee, which boils violently on contact. The technique is High confidence; the effect is smoky, caramelized, theatrically boiling. Vendor claims that the charcoal neutralizes acidity or aids digestion are medically unsubstantiated and are excluded from any quantitative claim (Observed, B, technique only).

Kumda kahve (Turkey). The cezve is buried in heated sand rather than set over flame. Sand gives even, controllable heat below a rolling boil, stabilizing the foam (kopuk) that marks quality; extraction is gentler and more even than direct-flame boiling (Observed, B).

6.4 Spice-roasted and spiced coffees

Cafe Touba (Senegal). Guinea pepper (djar, *Xylopia aethiopica*) and cloves are ground with the beans, and the brew is aerated by pouring from height. The pepper adds warming pungency over a

dark, heavy cup; the high pour aerates and cools (OBSERVED, B). The attribution of the recipe to a specific religious figure is Medium confidence.

Qahwa ma'atra (Morocco). Cinnamon, cardamom, and pepper ground with the beans give a warming, rounded spice cup (OBSERVED, C).

Cafe de olla (Mexico). Coffee boils in an unglazed clay pot with cinnamon and piloncillo (unrefined cane sugar). Boiling the sugar in the water gives homogeneous sweetness that stirred-in sugar cannot match; the clay pot adds slight mineral earthiness and the cinnamon integrates warm spice (OBSERVED, B). The pre-Hispanic origin story is Medium confidence.

6.5 Grain, root, and chicory extenders

Oliang (Thailand). Robusta is blended with roasted grains and seeds (corn, soy, sesame, rice) and brewed through a sock. The extenders dilute coffee intensity, adding smoky, malty body at low cost; caffeine per gram of powder falls proportionally (OBSERVED, B).

Chicory coffee (New Orleans; South Indian filter). Roasted chicory root blended with coffee, 10 to 30 percent in South Indian decoction. Chicory adds body and earthy bitterness, slows percolation, and dilutes caffeine since it contains none (OBSERVED, B).

Camp Coffee (Britain, 1885). A bottled coffee-chicory-sugar concentrate, the industrial endpoint of the extender logic: sweet, low-aroma coffee milk from a bottle (OBSERVED, B).

6.6 Leaf and cherry teas

Qishr (Yemen). Dried coffee cherry husks simmered with ginger and cardamom, drunk as a tea. Fruity, tea-like, near-decaffeinated since the husk carries little caffeine; one vendor source claims it is more widely drunk than coffee in Yemen, attributed as such (OBSERVED, B).

Kuti / kawa daun (Ethiopia / West Sumatra). Roasted coffee leaves simmered, salted in the Ethiopian kuti version. A smoky, caramel, low-caffeine leaf tea, chemically closer to tea than to coffee (OBSERVED, B).

6.7 Concentrates, layers, and foam architecture

Yuenyeung (Hong Kong). Three parts Hong Kong milk tea to seven parts coffee, smoothed with evaporated milk. The blend sits between coffee and tea; milk protein and fat round both beverages' edges (OBSERVED, B).

Bicerin (Turin). Layered hot chocolate, espresso, and cream in a small glass: strata of temperature, density, and flavor drunk in sequence (OBSERVED, B). *Marocchino* (Turin, 1980s) is the dry miniature: espresso, milk foam, cocoa dust in a demitasse (OBSERVED, C).

Cafe cubano. The first drops of the moka-style brew are whipped with sugar into espumita, a sweet crema-like foam spooned over the cups; the sugar foam masks bitterness while aerating the cup (OBSERVED, B).

Cafezinho (Brazil). Sugar boils in the water before the coffee is added, then the brew passes through a cloth passador. Pre-dissolved sugar gives homogeneous sweetness in a strong small cup (OBSERVED, B).

Kopi tubruk (Indonesia). Fine coffee immersed in the glass, grounds settling as you drink: heavy, gritty, the body intensifying toward the bottom (Observed, B).

6.8 Fire, embers, and historical boiling

Cafe tropeiro (Brazil). On the muleteer trails, grounds were boiled and settled by touching an ember-tipped stick to the brew. Field-expedient and unfiltered (Observed, C).

Ottoman boiling (Europe, 17th century). Flour-fine grounds decocted in the cezve: thick, aggressive extraction, the ancestor of Turkish method (McGowan, 2010) (Observed, B).

6.9 Animal-processed coffee

Kopi luwak (Indonesia). Civet digestion plus cherry selection. The quality effect decomposes into two parts: the animal selects the ripest cherries (a selection effect any careful picker replicates), and gut fermentation alters bean chemistry modestly. "Digestion makes it better" conflates the two (Observed, B).

Table 5. Twenty-six verified obscure, regional, and historical preparations

Method	Region	Key technique	Sensory effect	Conf.
Egg coffee	US Midwest	whole egg boiled with grounds	clear, low-bitterness	High
Ca phe trung	Hanoi	whipped yolk + condensed milk	custard foam over bitter base	High
Ca phe muoi	Hue	salted cream over iced robusta	salted-caramel rounding	High
Kopi tubruk	Indonesia	unfiltered immersion in glass	heavy, gritty, intensifying	High
Kopi joss	Yogyakarta	red-hot charcoal plunge	smoky, caramelized	High[a]
Hainanese kopi	Malaysia/Singapore	margarine+sugar wok roast; sock	buttery, caramelized, full	High
Oliang	Thailand	robusta + roasted grains; sock	smoky, malty, diluted	High
Yuenyeung	Hong Kong	3:7 coffee to milk tea	between coffee and tea	High
Cafe de olla	Mexico	clay pot, cinnamon, piloncillo	integrated sweet spice	High[b]
Cafezinho	Brazil	sugar boiled in water; cloth filter	homogeneous sweet, strong	High
Cafe tropeiro	Brazil	ember-tipped stick settling	field unfiltered cup	Med
Cafe cubano	Cuba	espumita: first drops + sugar	sweet foam, masked bitter	High
Bicerin	Turin	layered chocolate/espresso/cream	strata of flavor	High
Marocchino	Turin	espresso + foam + cocoa dust	dry chocolate mini	Med
Cafe Touba	Senegal	djar pepper + cloves ground in	warming spice heat	High[b]
Jebena buna	Ethiopia/Eritrea	clay pot; three re-brews	declining-strength ritual	High
Qishr	Yemen	cherry husks + ginger/cardamom	fruity tea, near-decaf	High
Kuti / kawa daun	Ethiopia/Sumatra	roasted leaves simmered	smoky, low-caffeine	High
Kumda kahve	Turkey	cezve in heated sand	even sub-boil, stable foam	High
Qahwa (dallah)	Arabia/Gulf	light roast + cardamom, long boil	floral, golden, bitter-sweet	High
Qahwa ma'atra	Morocco	spices ground with beans	warming rounded spice	Med
Kopi luwak	Indonesia	civet digestion + selection	smooth; partly selection	High
Kaffeost	Lapland	cheese cubes under coffee	two-phase, dairy-rounded	Med
Chicory coffee	New Orleans	roasted chicory blended in	fuller body, less caffeine	High
Camp Coffee	Britain	bottled coffee-chicory concentrate	sweet coffee milk	High
Ottoman boiling	Europe, 17th c.	flour-fine decoction in cezve	thick, aggressive	High

Notes: [a]Technique High; vendor health claims Low and excluded. [b]Method High; origin attribution Medium. Confidence definitions: High = multiple independent credible sources; Medium = one credible source or plausible secondary accounts; Low = single anecdote (none in this table; see Appendix C).
Source: Research notes, Section 6.

The mechanisms repeat across continents: clarification by protein (egg), bitterness suppression by salt, aroma carrying by fat, dilution and body from extenders, even low heat from sand, and selection effects mistaken for process effects (luwak). Obscurity does not mean inefficacy; several of these techniques solve real extraction problems that modern brewing rediscovers.

7 Water, grind, freshness, and sensory evaluation

7.1 Water chemistry

The SCA water standard specifies TDS 75 to 250 ppm (target about 150), calcium hardness 50 to 175 ppm as CaCO3, alkalinity 40 to 70 ppm as CaCO3, pH 6.5 to 7.5, and zero chlorine (Observed, A/B). Each parameter has a distinct role, and the literature recently corrected a popular misunderstanding.

Hendon et al. (2015) showed that magnesium binds flavor compounds more strongly than calcium, so magnesium-rich water extracts more flavor and caffeine, while high bicarbonate was associated with bitterness and sodium-softened water did not help extraction (Observed, A). A 2025 comprehensive review in the *Journal of Food Science* substantially qualified this: at realistic brewing concentrations, varying calcium and magnesium alone produced no detectable sensory difference, while alkalinity does the audible work (Journal of Food Science review, 2025) (Observed, A). The reconciliation: Hendon's binding-energy differences are real chemistry, but at brew-water concentrations the sensory effect of Ca/Mg variation is small, while bicarbonate's buffering of coffee acids is large and consistently perceptible in cup tests. Most very hard waters are also high in bicarbonate, and it is the bicarbonate that mutes acidity.

Distilled water near 0 ppm TDS has no hardness cations and no buffer: fewer dissolved ions means less favorable partitioning of flavor compounds into solution and no buffering of the coffee's own acids, yielding a cup that tastes simultaneously sharp and hollow (Observed, A/B). The SCA floor of 75 ppm exists for this reason.

7.2 Grind consistency

Particle size distribution, not just mean size, governs extraction uniformity. Fines below about 100 microns extract far faster than the mean particle, so a grind with many fines brews as a blend of over-extracted fines and under-extracted boulders (Observed, B). Uman et al. (2016) showed that grinding cold beans (below about -19 C) narrows the distribution and produces a more uniform fines peak around 10 microns, because cold beans fracture in a more brittle mode (Observed, A). Burr grinders produce a roughly log-normal distribution; blade grinders produce a wide bimodal distribution with excessive fines and boulders, which is why blade-ground coffee brews unevenly regardless of method (Observed, B).

7.3 Freshness, degassing, and storage

Roasting generates large volumes of CO2 trapped in the bean matrix, mostly via Strecker degradation. Whole beans release CO2 over about 30 days; grinding releases about 70 percent immediately (Illy and Viani, 1995) (Observed, B). Excess CO2 repels brew water from particle surfaces (the visible

bloom) and impedes even extraction: freshly roasted high-CO2 samples taste more sour, matured lower-CO2 samples more balanced, with sensory-relevant CO2 decline measurable out to 145 days (Observed, A).

Practical resting windows are trade consensus (Assumed, C) consistent with the degassing measurements: dark roasts 1 to 3 days, medium 3 to 7, light up to 14; espresso 5 to 14 days since CO2 disrupts puck saturation. Staling runs through volatile loss, lipid oxidation (papery, cardboard notes), and CO2 loss; grinding accelerates all of it by orders of magnitude of surface area. Best practice is whole beans in an airtight opaque container at cool room temperature, consumed within 2 to 4 weeks of roast, ground immediately before brewing (Observed, B/C).

7.4 Sensory science and grading

The 2004 SCAA cupping form scored ten categories to 100 points, with 80-plus defining specialty grade. In 2024 the SCA adopted the Coffee Value Assessment, splitting evaluation into descriptive assessment (what the coffee tastes like) and affective assessment (how much the cupper likes it), with the affective form retaining 100-point scoring, adding Sweetness and replacing Body with Mouthfeel (Observed, A). The split is the institutional recognition of the objective-perception gap: intensity is measurable and describable, preference is not reducible to it.

The Coffee Taster's Flavor Wheel (revised 2016 by SCA with World Coffee Research) organizes about 110 descriptors as a shared contracting lexicon, not a quality scale (Observed, B). The objective anchors: total solids correlate with body, headspace intensity with aroma, and caffeine and chlorogenic acid concentrations with bitterness and astringency (Gloess et al., 2013) (Observed, A).

8 Quantitative synthesis and misconception checks

8.1 Caffeine accounting

Caffeine in the cup is, to first order, dose times bean caffeine fraction times extraction fraction. The species bands (9 to 13 mg/g arabica, 15 to 25 mg/g canephora) set the input; roast level barely moves it per bean; the brew ratio and serving size set the output. Figure 6 shows the result: roughly 30 mg in a finjan of qahwa, roughly 300 mg in a phin of robusta, a tenfold range across methods from the same plant family.

Two decompositions are worth stating plainly. First, within one method, switching from arabica to robusta roughly doubles caffeine: the phin's 280 to 330 mg is mostly a robusta effect at a 1:8 to 1:10 ratio. Second, within one bean, switching methods moves caffeine by roughly a factor of five through ratio and serving size: a double espresso (120 to 190 mg) against a 320 ml pour-over (180 to 210 mg) against a 60 ml Turkish (50 to 65 mg).

8.2 Four misconceptions, corrected

1. "Espresso has more caffeine than drip." Per unit volume, true: Ludwig et al. (2014) measured 174 mg in 22 ml of espresso against 33 mg in 22 ml of filter coffee from the same beans (Observed,

A). Per typical serving, usually false: a 1 oz single carries about 63 mg against about 95 mg in 8 oz of drip (USDA, 2019) (Observed, A/B). Serving size dominates; concentration misleads.

2. "Robusta is just bad arabica with more caffeine." The caffeine clause is correct (roughly double). The quality clause is not: the gap is agronomic and processing-driven, fine robusta can score at specialty levels, and caffeine's own bitterness is part of why robusta tastes harsher (Observed, B). The error is the inference from "more caffeine" to "better" or from "robusta" to "defective."

3. "Dark roast has more caffeine." False as usually meant. By weight, roast degree has no significant effect on caffeine content (Observed, A); by scoop, light roast delivers slightly more because dark beans are less dense (Reconstructed, B/C). Dark roast tastes stronger because roasting manufactures bitter compounds (Section 4.3), not because it concentrates caffeine.

4. "Strong-tasting coffee has more caffeine." Perceived strength tracks TDS and bitter roast compounds, not caffeine. Decaf tastes "strong" at the same TDS; caffeine contributes only about 15 percent of perceived bitterness (Observed, A/B). The variables that actually raise caffeine per serving are dose, robusta share, and low brew ratio, in that order; roast level and perceived intensity sit near the bottom of the list.

9 Discussion: falsifiability and limitations

This is a retrospective synthesis; there was no pre-registered plan. The paper's load-bearing claims and what would overturn them:

- The species caffeine bands (arabica 9–13 mg/g, canephora 15–25 mg/g) would be falsified by a multi-origin, multi-crop analytical survey showing systematically different bands; the existing corroboration across independent labs makes this unlikely but the liberica band is genuinely unresolved.
- The claim that roasting does not meaningfully change per-bean caffeine would be falsified by controlled roast-degree series showing large per-bean losses; Rao et al. (2024) is the current best evidence and it supports stability.
- The brewing control chart positions would be falsified by published refractometer data on the traditional methods; we predict Turkish, phin, and qahwa will land near their estimated positions, and welcome the measurements.
- The processing directionality (washed bright, natural sweet and heavy, wet-hulled low-acid) would be falsified by controlled same-lot trials reversing the ranking; the metabolomics trial supports the current ordering.

Limitations are stated plainly. Roast-level definitions are not standardized across studies, so the CGA retention bands in Table 2 are indicative, not calibrated. Traditional brewing methods are under-instrumented: six of fourteen methods in Tables 3 and 4 lack published TDS or EY data, and their caffeine figures are arithmetic estimates. Liberica and Excelsa chemistry is thin-data. The 2025 liberica taxonomy revision postdates most cited work. Between-study variation in absolute caffeine values is substantial (for example, Krol et al. (2020) reported absolute levels far below other labs, suggesting a method issue), so cross-study comparisons should use within-study ratios where possible. The obscure-methods section relies partly on journalism and trade sources where ethnographic literature does not exist in English; confidence ratings mark the boundary.

10 Conclusion: an evidence-based exploration guide

The paper's quantitative spine supports a short list of contrasts worth tasting, each chosen to isolate one variable the data show to matter.

Species. Cup a washed Ethiopian arabica against a fine robusta (Vietnamese or Indian) at the same roast and brew: the experiment isolates the sucrose-lipid-caffeine chemistry of Table 1. Then try a liberica or Excelsa, the third species the market forgot, for the woody, jackfruit, savory-sweet profile no arabica produces.

Processing. Buy one lot processed two ways (washed versus natural, increasingly sold as pairs): the cleanest controlled experiment available to a drinker, and the direction in Figure 3 predicts the outcome. Then try an anaerobic or carbonic maceration lot to taste controlled fermentation at its current frontier.

Roast. Cup the same coffee at light and dark roast side by side: acidity and origin character against bitterness and body, with caffeine held roughly constant per bean. Weigh the doses; the per-scoop difference is the density lesson made tangible.

Brewing. Make espresso and moka from the same coffee to feel the extraction-efficiency gap; brew V60 and French press side by side for the clarity-body tradeoff at matched strength; try a phin with robusta and condensed milk for the highest-caffeine mainstream serving, and a finjan of cardamom qahwa for the lowest.

Obscure. Three entries reward the effort with genuinely different mechanisms: Scandinavian-American egg coffee (protein clarification of a boiled brew), kopi joss (charcoal-fired theater with the technique separated from its health claims), and qishr (the cherry tea Yemen drinks more than coffee). Each changes the cup through a mechanism no mainstream method uses.

The throughline of the evidence is simple: the cup is set first by species chemistry, then by roast chemistry, then by the arithmetic of dose, ratio, and serving size. Everything else, method mystique included, is commentary on those three.

A Data and methods notes

All figures were produced from the CSV files in the paper's data directory, whose README records sources and data-through dates. Figure 4's caffeine-per-gram index is arithmetic from mass loss (caffeine conserved, mass lost 12 to 22 percent), not a new measurement. Figure 3's 1-to-5 scores are reconstructed from the qualitative comparison table in the research notes and are graded C. Caffeine-per-serving arithmetic uses 1.2 percent caffeine by dry weight for arabica and 2.2 percent for robusta unless a study reported its own bean values, with extraction fractions stated in the text.

B Robustness checks

Species bands. The arabica and canephora caffeine bands are corroborated by three independent sources (Farah 2012; Foods 2020 species-discrimination study; Brazilian commercial samples) at the same roughly 2:1 ratio. The sucrose disagreement (Farah 0.9–4.0 percent versus Brunning 2023 3.0–7.0 percent for robusta) is reported, not resolved; the paper's conclusions do not depend on it.

Roast caffeine stability. More than twenty studies agree that per-bean caffeine is essentially unaffected by roast degree; the outliers (e.g., Krol et al. (2020)) show absolute levels inconsistent with other labs and are treated as method issues.

Water chemistry. The paper reports both Hendon et al. (2015) and the 2025 *Journal of Food Science* review, with the review as the more current authority.

Obscure methods. Entries required named sources; six leads that could not be verified are quarantined in Appendix C and excluded from all claims.

C Unverified leads

The following were investigated and excluded for lack of credible sourcing: a Turkish bride's salt-coffee custom (single travel anecdote); Ethiopian buttered coffee as a standalone preparation (only variant notes found); deliberately fermented indigenous coffee beverages beyond kopi luwak (no credible source); "coffee cooked over open fire" as a named single-town method (covered in substance by cafe de olla, jebena, and Ottoman boiling); detailed attribution of ca phe muoi to a specific 2010 cafe (single guide's reporting); kopi joss vendor health claims (medically unsubstantiated).

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